

Physics of the Neutron Star Spectral Image Cube Pipeline: A Code–Paper Cross-Reference Report

NeutronStar.jl Documentation

Auto-generated from source code review

24 March 2026

Abstract

This report documents every physics equation implemented in the `NeutronStar.jl` pipeline, from the equation of state and TOV stellar structure through magnetic radiative transfer opacities and atmosphere solvers to the general-relativistic ray tracing and colorimetric rendering. Each equation is cross-referenced against its source publication with explicit equation numbers. Discrepancies between the code and the published formulae are flagged.

Contents

1	Neutron Star Structure	4
1.1	Physical constants	4
1.2	BSk equation of state	4
1.2.1	Pressure–density relation	4
1.2.2	Fit parameters	5
1.2.3	Energy density	5
1.2.4	Pseudo-enthalpy	6
1.2.5	Density inversion	6
1.3	TOV equations	6
1.3.1	Central boundary conditions	7
1.3.2	Numerical integration	7
1.3.3	Derived quantities	8
1.3.4	Mass–radius curve	8
1.4	Summary of approximations	8
1.5	Summary of discrepancies	9
2	Opacities	10
2.1	Non-magnetic opacities	10
2.1.1	Free-free absorption	10
2.1.2	Thomson scattering	10
2.1.3	Total opacity and scattering albedo	10
2.1.4	Non-magnetic Rosseland mean	10
2.2	Magnetic opacities: basic polarisation cross-sections	11
2.2.1	Magnetic electron scattering	11
2.2.2	Electron free-free absorption	11
2.2.3	Proton-proton collisions	12
2.2.4	Total polarisation cross-section	12
2.3	Coulomb logarithms	12
2.3.1	Classical (non-magnetic) Coulomb logarithm	12

2.3.2	Magnetic Coulomb logarithm	12
2.3.3	Proton-proton Coulomb logarithm	13
2.4	Dielectric tensor and polarisation weights	13
2.5	Normal-mode opacities and the absorption/scattering split	14
2.6	Angle-averaged opacities and Rosseland means	14
2.6.1	Magnetic Rosseland means	15
2.7	Summary of equation cross-references	15
2.8	Discrepancies and implementation notes	15
3	Radiative Transfer	17
3.1	Atmosphere Structure	17
3.1.1	Hydrostatic equilibrium and equation of state	17
3.1.2	Eddington temperature profile	17
3.2	Feautrier Method	18
3.2.1	Second-order transfer equation	18
3.2.2	Block-tridiagonal system	18
3.2.3	Boundary conditions	19
3.2.4	Forward elimination and back-substitution	19
3.3	Frequency-Adaptive Depth Grid	19
3.4	Rybicki Temperature Correction	20
3.4.1	Linearised moment equation	20
3.4.2	Tridiagonal system per frequency	20
3.4.3	Global $\bar{J}$ system	21
3.4.4	Boundary conditions for the Rybicki system	21
3.5	Two-Mode Magnetic Extension	21
3.5.1	Per-mode transfer equation	21
3.5.2	Two-mode Rybicki correction	22
3.5.3	Uniform flux correction	22
3.6	Convergence Criteria and Iteration Scheme	23
4	Surface Model and Dipole Geometry	24
4.1	Magnetic colatitude	24
4.2	Greenstein–Hartke temperature map	24
4.3	Dipole magnetic field strength	24
5	Schwarzschild Geodesic Ray Tracing	24
5.1	The cosine relation	25
5.2	Impact parameter	25
5.3	Exact bending-angle integral	25
5.4	Visible fraction of the stellar surface	25
5.5	Image construction	26
6	Relativistic Intensity Transform	26
7	Spectral Image Cube and Atmosphere Grid Lookup	27
7.1	Pre-computed atmosphere grid	27
7.2	Grid interpolation	27
7.3	Per-pixel spectrum computation	27

8	CIE 1931 Colorimetry and sRGB Rendering	28
8.1	CIE 1931 XYZ tristimulus values	28
8.2	XYZ to linear sRGB	28
8.3	Tone mapping	28
8.4	sRGB gamma correction	28
8.5	False-colour X-ray rendering	29

1 Neutron Star Structure

This section documents Module 1 of the visualization pipeline: the computation of static, spherically symmetric neutron star models via the Tolman–Oppenheimer–Volkoff (TOV) equations, closed by the Brussels–Montreal BSk unified equations of state. Every equation implemented in the code is stated below, cross-referenced against the source literature, and any approximations or discrepancies are flagged.

1.1 Physical constants

All physical constants are taken in CGS units from the CODATA 2018 adjustment [?](#). The values used are:

$$c = 2.997\,924\,58 \times 10^{10} \text{ cm s}^{-1}, \quad G = 6.674\,30 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}. \quad (1)$$

The solar mass is set to $M_\odot = 1.988\,92 \times 10^{33} \text{ g}$, attributed to IAU 2015 nominal values.

Remark. The IAU 2015 nominal solar mass parameter is $\mathcal{GM}_\odot^N = 1.327\,124\,4 \times 10^{26} \text{ cm}^3 \text{ s}^{-2}$ [?](#), which combined with the CODATA 2018 value of G yields $M_\odot \approx 1.988\,48 \times 10^{33} \text{ g}$. The code value $1.988\,92 \times 10^{33} \text{ g}$ is a commonly used older estimate and differs by $\sim 0.02\%$. This has negligible impact on computed stellar masses at the precision of the EOS fit ($\sim 1\%$).

1.2 BSk equation of state

The equation of state (EOS) is an analytical representation of the unified Brussels–Montreal BSk models, following Potekhin et al. (2013) [?](#) (hereafter P13). The code implements three parameter sets—BSk19, BSk20, and BSk21—with BSk21 used as the default (stiffest EOS, maximum mass $M_{\text{max}} \approx 2.27 M_\odot$).

1.2.1 Pressure–density relation

Introducing the dimensionless logarithmic variables

$$\xi \equiv \log_{10}(\rho/\text{g cm}^{-3}), \quad \zeta \equiv \log_{10}(P/\text{dyn cm}^{-2}), \quad (2)$$

the analytical fit for $P(\rho)$ is given by P13, Eq. (3):

$$\begin{aligned} \zeta = & \frac{a_1 + a_2 \xi + a_3 \xi^3}{1 + a_4 \xi} \left\{ \exp[a_5(\xi - a_6)] + 1 \right\}^{-1} \\ & + (a_7 + a_8 \xi) \left\{ \exp[a_9(a_6 - \xi)] + 1 \right\}^{-1} \\ & + (a_{10} + a_{11} \xi) \left\{ \exp[a_{12}(a_{13} - \xi)] + 1 \right\}^{-1} \\ & + (a_{14} + a_{15} \xi) \left\{ \exp[a_{16}(a_{17} - \xi)] + 1 \right\}^{-1} \\ & + \frac{a_{18}}{1 + [a_{19}(\xi - a_{20})]^2} + \frac{a_{21}}{1 + [a_{22}(\xi - a_{23})]^2}. \end{aligned} \quad (3)$$

The pressure is then $P = 10^\zeta$.

Verification against paper. The implementation in `bsk_eos.jl` (lines 121–127) reproduces Eq. (3) term-by-term with no discrepancies. Each of the six additive terms matches P13 Eq. (3) identically: four Fermi–Dirac-type terms and two Lorentzian terms.

Validity range. The fit is valid for $10^6 \lesssim \rho \lesssim 10^{16} \text{ g cm}^{-3}$ ($6 \lesssim \xi \lesssim 16$). The code enforces $\rho > 0$ and $\rho < 10^{17} \text{ g cm}^{-3}$ via assertions. Per P13, the typical fit error on P is $\approx 1\%$ for $\xi \gtrsim 6$, with a maximum of 3.7% at $\xi = 9.51$ (the neutron-drip transition region).

1.2.2 Fit parameters

The 23 parameters $a_1, \dots, a_{23}$ for each BSk model are taken from P13 Table 2. All parameter values in the code have been checked against the published table; the comparison for the preferred BSk21 model is shown in Table 1.

Table 1: BSk21 parameters a_i : code values versus P13 Table 2. All entries match exactly to the published precision.

i	Code	P13 Table 2
1	4.857	4.857
2	6.981	6.981
3	0.00706	0.00706
4	0.19351	0.19351
5	4.085	4.085
6	12.065	12.065
7	10.521	10.521
8	1.5905	1.5905
9	4.104	4.104
10	-28.726	-28.726
11	2.0845	2.0845
12	4.89	4.89
13	14.302	14.302
14	22.881	22.881
15	-1.7690	-1.7690
16	0.989	0.989
17	15.313	15.313
18	0.091	0.091
19	4.68	4.68
20	11.65	11.65
21	-0.086	-0.086
22	10.0	10.0
23	14.15	14.15

The same check was performed for BSk19 and BSk20. All $3 \times 23 = 69$ parameter values match the published table to every quoted digit. No transcription errors were found.

1.2.3 Energy density

For the TOV system, the total energy density ε is required. Following P13 Sect. 3.1 (discussion below their Eq. 1), the variable ρ in the BSk parametrization is the *gravitational mass density*, which already incorporates internal (binding) energy contributions. The code therefore uses

$$\varepsilon = \rho c^2, \quad (4)$$

which is the correct identification for this parametrization.

Approximation. This relation is exact within the P13 fitting framework. In a more fundamental treatment, one would distinguish rest-mass energy density $\rho_0 c^2$ from thermal and interaction

contributions. The BSk fit absorbs these corrections into the definition of ρ , so Eq. (4) is self-consistent.

1.2.4 Pseudo-enthalpy

The pseudo-enthalpy, useful for rotating-star calculations, is defined by P13 Eq. (2):

$$H = \int_0^P \frac{dP'}{\rho(P') c^2 + P'} . \quad (5)$$

The code evaluates this integral via the trapezoidal rule in $\log_{10} P$ space with $N = 1000$ points (default), using the change of variables $dP' = P' \ln 10 d(\log_{10} P')$, so that the integrand becomes

$$H = \int_{\log_{10} P_{\min}}^{\log_{10} P} \frac{P' \ln 10}{\varepsilon(P') + P'} d(\log_{10} P') , \quad (6)$$

where $P_{\min} = P(\rho = 10^6 \text{ g cm}^{-3})$ is the lower limit of the EOS fit validity.

Verification against paper. Equation (5) matches P13 Eq. (2) identically. The logarithmic integration variable is a numerical convenience not present in the paper; it improves quadrature accuracy over the many decades of pressure spanning the stellar interior.

Approximation. The function $\rho(P)$ required inside the integrand is obtained by bisection inversion of $P(\rho)$ (see Eq. 3), with relative tolerance 10^{-10} in $\log_{10} \rho$. The trapezoidal rule with $N = 1000$ uniform steps in $\log_{10} P$ yields accuracy well within the underlying $\sim 1\%$ EOS fit error.

1.2.5 Density inversion

Since the BSk fit provides $P(\rho)$ analytically but $\rho(P)$ is needed for the TOV integration, the inverse function is computed by bisection in $\log_{10} \rho$ space:

$$\text{find } \rho^* \text{ such that } P(\rho^*) = P_{\text{target}}, \quad \rho^* \in [10^4, 10^{16}] \text{ g cm}^{-3}. \quad (7)$$

The bracket is automatically extended by factors of 10 if the target pressure falls outside the initial range. Convergence to relative tolerance 10^{-10} is guaranteed within 200 iterations ($2^{200} \gg 10^{12}$, the bracket ratio in decades).

Approximation. Bisection is unconditionally stable but slower than Newton–Raphson. Since $dP/d\rho$ is not available analytically from the fit (it would require differentiating Eq. 3), bisection is the simplest robust choice. The ~ 40 iterations typically needed add negligible cost compared to the RK4 TOV loop.

1.3 TOV equations

The structure of a static, spherically symmetric star in general relativity is governed by the TOV equations ????. Following the notation of Shapiro & Teukolsky (1983), Ch. 5–6:

$$\frac{dP}{dr} = - \frac{G(\varepsilon + P) \left(m + \frac{4\pi r^3 P}{c^2} \right)}{r(r c^2 - 2 G m)} , \quad (8)$$

$$\frac{dm}{dr} = \frac{4\pi r^2 \varepsilon}{c^2} , \quad (9)$$

where r is the circumferential (areal) radius, $m(r)$ is the enclosed gravitational mass, $P(r)$ is the pressure, and $\varepsilon(r) = \rho(r) c^2$ is the total energy density. The system is closed by the EOS $P = P(\rho)$ from Sect. 1.2.

Verification against paper. The code implementation in `tov.jl` (lines 167–175) reproduces Eqs. (8)–(9) exactly. The function `tov_rhs` returns $(dm/dr, dP/dr)$ with the signs and grouping matching the standard form. An assertion checks that the denominator $r(rc^2 - 2Gm) > 0$, i.e. that the integration has not crossed the Schwarzschild horizon.

1.3.1 Central boundary conditions

At $r = 0$ the TOV equations are singular (0/0 form). The code avoids this by starting the integration at a small radius r_0 using a Taylor expansion about the centre. To derive these, one substitutes $m(r) = (4\pi/3)(\varepsilon_c/c^2)r^3 + \mathcal{O}(r^5)$ into Eq. (8) and expands to leading non-trivial order:

$$m(r_0) = \frac{4\pi}{3} \frac{\varepsilon_c}{c^2} r_0^3, \quad (10)$$

$$P(r_0) = P_c - \frac{2\pi G}{3c^4} (\varepsilon_c + P_c)(\varepsilon_c + 3P_c) r_0^2. \quad (11)$$

These are the standard results (see, e.g., Shapiro & Teukolsky 1983, Eq. (5.7.6); Lindblom 1992 ?).

Verification against code. The code (`tov.jl`, lines 73–75) implements:

```
m0 = 4pi/3 * eps_c / c^2 * r0^3
dP_coeff = -2pi * G * (eps_c + P_c) * (eps_c + 3P_c) / (3 * c^4)
P0 = P_c + dP_coeff * r0^2
```

This matches Eqs. (10)–(11) exactly. The default starting radius is $r_0 = 100 \text{ cm} = 1 \text{ m}$, which is negligible compared to stellar radii of order 10^6 cm and ensures the $\mathcal{O}(r^4)$ truncation error is of order $(r_0/R)^4 \sim 10^{-16}$.

1.3.2 Numerical integration

The TOV system is integrated outward using a classical fourth-order Runge–Kutta (RK4) scheme. The step size Δr is adaptive:

$$\Delta r = \min\left(\Delta r_{\max}, \Delta r_{\text{init}} \cdot \max(1, (P/P_{\text{surf}})^{0.3}), 0.01 r\right), \quad (12)$$

with defaults $\Delta r_{\text{init}} = 100 \text{ cm}$, $\Delta r_{\max} = 1000 \text{ cm}$. The three constraints ensure that steps are (i) bounded above, (ii) small near the surface where P changes rapidly, and (iii) never exceed 1% of the current radius.

Approximation. The step controller in Eq. (12) is heuristic rather than based on local truncation error estimation. It does not use embedded Runge–Kutta pairs (e.g. Dormand–Prince) for rigorous error control. However, the small maximum step size ($\Delta r_{\max} = 10 \text{ m}$, compared to $R \sim 10 \text{ km}$) and the additional 1%-of- r constraint ensure that the integration is well resolved.

Surface termination. Integration terminates when P drops below the surface pressure $P_{\text{surf}} \equiv P(\rho = 10^6 \text{ g cm}^{-3})$, which corresponds to the lower validity limit of the BSk fit. At the final step, a linear interpolation in r locates the exact surface crossing:

$$r_{\text{surf}} = r_{n-1} + \frac{P_{n-1} - P_{\text{surf}}}{P_{n-1} - P_n} \Delta r, \quad (13)$$

and the mass is interpolated likewise.

1.3.3 Derived quantities

From the integration output—the total gravitational mass $M \equiv m(R)$ and circumferential radius $R \equiv r_{\text{surf}}$ —the code computes the following:

Compactness.

$$u \equiv \frac{2 G M}{R c^2}. \quad (14)$$

This is the standard relativistic compactness parameter (the ratio of Schwarzschild radius to stellar radius).

Surface gravity.

$$g_{\text{surf}} = \frac{G M}{R^2 \sqrt{1-u}} = \frac{G M}{R^2} \left(1 - \frac{2 G M}{R c^2}\right)^{-1/2}. \quad (15)$$

This is the *local* surface gravity, *not* the redshifted gravity $g_{\infty} = g_{\text{surf}} \sqrt{1-u} = GM/R^2$ seen by a distant observer. The code comment labels it “redshifted”, but the formula actually gives the local value—the redshifted surface gravity would instead be $g_{\infty} = GM/R^2$.

Discrepancy. *The code comment on line 184 of `tov.jl` describes `g_surf` as “surface gravity (redshifted)”, but the implemented formula $g = GM/(R^2 \sqrt{1-2GM/Rc^2})$ is the local (proper) surface gravity, not the redshifted one. The formula itself is correct for the local surface gravity; only the comment is misleading.*

1.3.4 Mass–radius curve

The function `mass_radius_curve` varies the central density logarithmically over $\log_{10}(\rho_c/\text{g cm}^{-3}) \in [14.0, 15.8]$ and calls `solve_tov` at each value to trace the $M(R)$ relation. This range spans from $\rho_c \approx 10^{14} \text{ g cm}^{-3}$ (slightly below nuclear saturation) to $\rho_c \approx 6.3 \times 10^{15} \text{ g cm}^{-3}$ (well above the maximum-mass configuration).

1.4 Summary of approximations

The key approximations and simplifications in Module 1 are:

1. **EOS fit accuracy:** The BSk analytical fit (Eq. 3) reproduces the tabulated EOS to $\sim 1\%$ for $\xi \gtrsim 6$, with a worst case of 3.7% at $\xi = 9.51$ (P13, Sect. 3.2).
2. **Zero temperature:** The BSk parametrization assumes cold, catalyzed matter ($T = 0$). Thermal corrections are neglected.
3. **Spherical symmetry:** Rotation is neglected; the star is treated as static and spherically symmetric.
4. **Heuristic step control:** The RK4 integrator uses an empirical adaptive step (Eq. 12) rather than embedded error estimation.
5. **Solar mass value:** $M_{\odot}$ differs from the precise IAU 2015 nominal value by $\sim 0.02\%$ (Sect. 1.1).
6. **Density inversion by bisection:** $\rho(P)$ is obtained numerically rather than from a separate analytical inverse fit (P13 provide such fits in their Eqs. 6–7, which are not used here).

1.5 Summary of discrepancies

1. **Surface gravity comment (cosmetic):** The code comment on `tov.jl` line 184 labels `g_surf` as “redshifted”, but the formula computes the local proper surface gravity. The formula is correct; only the comment is wrong. See Eq. (15).
2. **Solar mass value (minor):** $M_{\odot} = 1.98892 \times 10^{33}$ g vs. the IAU 2015/CODATA 2018 derived value of 1.98848×10^{33} g (a 0.02% difference).
3. **No discrepancies in physics equations:** All implemented equations—the BSk pressure fit (Eq. 3), TOV system (Eqs. 8–9), central expansion (Eqs. 10–11), and pseudo-enthalpy (Eq. 5)—match the source references exactly. All 69 BSk fit parameters across three models match P13 Table 2 to every quoted digit.

2 Opacities

This module computes monochromatic and Rosseland-mean radiative opacities for fully ionised hydrogen, covering both the non-magnetic ($B = 0$) limit and the strongly magnetised regime where the field is quantising ($\beta_e \equiv \hbar\omega_{ce}/(k_B T) \gg 1$). The magnetic treatment follows Potekhin & Chabrier (2003, hereafter PC03)¹, cross-referenced by equation number below. The non-magnetic limit uses the Gaunt-factor-based formulae of Haakonsen et al. (2012)². Implementation resides in six source files under `src/opacity/`.

2.1 Non-magnetic opacities

2.1.1 Free-free absorption

The non-magnetic free-free absorption opacity per unit mass for fully ionised hydrogen is (`hydrogen_ff.jl`, function `kappa_ff`; Haakonsen et al. Eq. 12):

$$\kappa_\nu^{\text{ff}} = \frac{4e^6}{3m_e h c} \sqrt{\frac{2\pi}{3k_B m_e}} \frac{\rho}{(m_p + m_e)^2} \nu^{-3} (1 - e^{-u}) \tilde{g}_{\text{ff}}(u, \gamma^2), \quad (16)$$

where $u = h\nu/(k_B T)$ and $\gamma^2 = \text{Ry}/(k_B T)$ with $\text{Ry} = 13.6057$ eV. The thermally-averaged Gaunt factor $\tilde{g}_{\text{ff}}$ is obtained by bilinear interpolation on the Sutherland (1998) table (`gaunt_ff.jl`, struct `GauntTable`). Boundary values are clamped to the table edges, reproducing the McPHAC convention.

2.1.2 Thomson scattering

The reduced Thomson scattering opacity for fully ionised hydrogen (Haakonsen Eq. 11 with $f_{\text{ion}} = 1$) is

$$\sigma_{\text{T},t} = \frac{\sigma_T}{m_p + m_e}, \quad (17)$$

where $\sigma_T = 6.652 \times 10^{-25} \text{ cm}^2$ is the non-magnetic Thomson cross-section.

2.1.3 Total opacity and scattering albedo

The total extinction opacity is

$$k_\nu = \kappa_\nu^{\text{ff}} + \sigma_{\text{T},t}, \quad (18)$$

and the scattering albedo (Haakonsen Eq. 6) is $\varrho_\nu = \sigma_{\text{T},t}/k_\nu$.

2.1.4 Non-magnetic Rosseland mean

The Rosseland mean opacity (`hydrogen_ff.jl`, function `rosseland_mean`) is computed by numerical quadrature on a discrete frequency grid:

$$\frac{1}{\kappa_R} = \frac{\int_0^\infty \frac{1}{k_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}, \quad (19)$$

where the Planck derivative is

$$\frac{\partial B_\nu}{\partial T} = \frac{2h^2\nu^4}{c^2 k_B T^2} \frac{e^u}{(e^u - 1)^2}. \quad (20)$$

The integrals are evaluated by the midpoint rule on a log-spaced grid. A fallback to $\sigma_{\text{T},t}$ is returned when the denominator vanishes.

¹A. Y. Potekhin & G. Chabrier, *ApJ* **585**, 955 (2003).

²C. B. Haakonsen et al., *ApJ* **749**, 52 (2012).

2.2 Magnetic opacities: basic polarisation cross-sections

In a magnetised plasma, radiative transfer decomposes into three basic cyclic polarisations $\alpha \in \{-1, 0, +1\}$ (left-circular, longitudinal, right-circular with respect to $\mathbf{B}$). The code defines the electron and proton cyclotron frequencies as

$$\omega_{ce} = \frac{eB}{m_e c}, \quad \omega_{cp} = \frac{eB}{m_p c}, \quad (21)$$

and the quantisation parameter (PC03 Eq. 1)

$$\beta_e = \frac{\hbar \omega_{ce}}{k_B T}. \quad (22)$$

2.2.1 Magnetic electron scattering

The magnetic Thomson cross-section for polarisation α is (PC03 Eq. 33; `magnetic_ff.jl`, function `sigma_scatter_alpha`):

$$\sigma_{\alpha}^{s,e} = \frac{\omega^2}{(\omega + \alpha \omega_{ce})^2 + \nu_{e,\alpha}^2} \sigma_T, \quad (23)$$

where the natural (radiative) damping frequency is (PC03 Eq. 39)

$$\nu_e^s = \frac{2}{3} \frac{e^2}{m_e c^3} \omega^2. \quad (24)$$

In the non-magnetic limit ($B \rightarrow 0$) every polarisation receives $\sigma_T/3$.

2.2.2 Electron free-free absorption

The magnetic free-free cross-section per polarisation is (PC03 Eq. 51; `magnetic_ff.jl`, function `sigma_ff_alpha`):

$$\sigma_{\alpha}^{\text{ff}} \approx \frac{\omega^2}{(\omega + \alpha \omega_{ce})^2 (\omega - \alpha \omega_{cp})^2 + \omega^2 \tilde{\nu}_{\alpha}^2} \frac{4\pi e^2 \nu_{\alpha}^{\text{ff}}}{m_e c}, \quad (25)$$

where $\tilde{\nu}_{\alpha}$ is the combined damping frequency (PC03 Eq. 52):

$$\tilde{\nu}_{\alpha} = \nu_{\alpha}^{\text{ff}} + \left(1 + \alpha \frac{\omega_{ce}}{\omega}\right) \nu_p + \left(1 - \alpha \frac{\omega_{cp}}{\omega}\right) \nu_e, \quad (26)$$

and the effective free-free collision frequency is (PC03 Eq. 41):

$$\nu_{\alpha}^{\text{ff}} = \frac{4}{3} \sqrt{\frac{2\pi}{m_e k_B T}} \frac{n_e e^4}{\hbar \omega} (1 - e^{-u}) \Lambda_{\alpha}^{\text{ff}}, \quad u \equiv \frac{\hbar \omega}{k_B T}. \quad (27)$$

The factor $(1 - e^{-u})$ accounts for induced (stimulated) radiation.

Note on damping terms. In Eq. (26) the electron damping is $\nu_e = \nu_e^s + \nu_e^{\text{pp}}$ and the proton damping is $\nu_p = \nu_p^s + \nu_p^{\text{pp}}$, where $\nu_p^s = (2/3)(e^2/(m_p c^3))\omega^2$ is the proton radiative width (analogous to Eq. (24)). The code sets ν_e and ν_p from these sums before constructing $\tilde{\nu}_{\alpha}$. This is consistent with the structure of PC03 Eqs. 38–39 and 52, though PC03 do not write out an explicit “ ν_p ” symbol.

2.2.3 Proton-proton collisions

Absorption due to proton-proton collisions is given by (PC03 Eq. 46; `magnetic_ff.jl`, function `sigma_pp_alpha`):

$$\sigma_{\alpha}^{\text{pp}} = \frac{1}{(\omega - \alpha \omega_{\text{cp}})^2 + \nu_{p,\alpha}^2} \frac{4\pi e^2 \nu^{\text{pp}}}{m_p c}, \quad (28)$$

where the proton damping frequency for this channel is (PC03 Eq. 53):

$$\nu_{p,\alpha} = \frac{m_e}{m_p} \nu_{\alpha}^{\text{ff}} + \nu_p^s + \nu^{\text{pp}}, \quad (29)$$

and ν^{pp} is the effective proton-proton collision frequency (PC03 Eq. 47; `magnetic_ff.jl`, function `nu_pp`):

$$\nu^{\text{pp}} = \frac{256}{3} \sqrt{\frac{\pi}{m_p k_B T}} \frac{n_p e^4}{\hbar \omega} \frac{k_B T}{m_p c^2} (1 - e^{-u}) \Lambda_{\text{pp}}. \quad (30)$$

2.2.4 Total polarisation cross-section

For fully ionised hydrogen ($x_H = 0$), the total cross-section per polarisation is (`magnetic_ff.jl`, function `sigma_total_alpha`):

$$\sigma_{\alpha}^{\text{tot}} = \sigma_{\alpha}^{\text{ff}} + \sigma_{\alpha}^{\text{pp}} + \sigma_{\alpha}^{s,e}. \quad (31)$$

This omits proton scattering $\sigma_{\alpha}^{s,p}$ (PC03 Eq. 34), which scales as $(m_e/m_p)^2 \approx 3 \times 10^{-7}$ relative to $\sigma_{\alpha}^{s,e}$ and is negligible away from the proton cyclotron resonance.

2.3 Coulomb logarithms

2.3.1 Classical (non-magnetic) Coulomb logarithm

In the $B = 0$ limit, the thermally averaged Coulomb logarithm is (PC03 Eq. 43; `coulomb_magnetic.jl`, function `coulomb_log_classical_safe`):

$$\Lambda_{\text{cl}}^{\text{ff}} = e^{u/2} K_0(u/2), \quad (32)$$

where K_0 is the modified Bessel function of the second kind. For $u/2 > 300$ the code uses the asymptotic form $e^x K_0(x) \sim \sqrt{\pi/(2x)}$ to avoid overflow.

2.3.2 Magnetic Coulomb logarithm

When the field is quantising ($\beta_e > 0.1$), the Born-approximation Coulomb logarithm replaces $\Lambda_{\text{cl}}^{\text{ff}}$ (PC03 Eq. 44a; `coulomb_magnetic.jl`, function `coulomb_log_magnetic`):

$$\Lambda_{\alpha}^{\text{ff}} = \frac{3}{4} e^{u/2} \sum_{n=-\infty}^{\infty} \int_0^{\infty} Q_n^{\alpha}(\beta_e, u, y) dy. \quad (33)$$

The integrand Q_n^{α} is given by (PC03 Eqs. 44b–e):

$$Q_n^{\alpha} = \frac{y}{\zeta} [(y + \theta + \zeta) \sinh(\beta_e/2)]^{-|n|} A_n^{\alpha}, \quad (34)$$

$$A_n^0 = \frac{x_n K_1(x_n)}{y + \beta_e/4}, \quad (35)$$

$$A_n^{\pm 1} = \frac{y + \theta + |n| \zeta}{\zeta^2} K_0(x_n), \quad (36)$$

with auxiliary quantities (PC03 Eqs. 44d–e):

$$\zeta = \sqrt{1 + 2\theta y + y^2}, \quad \theta = \frac{1 + e^{-\beta_e}}{1 - e^{-\beta_e}}, \quad x_n = \frac{|u - n\beta_e|}{\sqrt{0.25 + y/\beta_e}}. \quad (37)$$

The Landau-level sum is truncated at $|n| \leq N_{\max} = \min(\max(2, \lceil 3/\beta_e + 2 \rceil), 50)$; for strongly quantising fields ($\beta_e \gg 1$) the $n = 0$ term dominates. Each integral is evaluated by adaptive quadrature (QuadGK) over $y \in [0, 20]$ with a relative tolerance of 10^{-4} .

2.3.3 Proton-proton Coulomb logarithm

The proton-proton Coulomb logarithm is the analytic fit (PC03 Eq. 48; `magnetic_ff.jl`, function `coulomb_log_proton`):

$$\Lambda_{pp} \approx 0.6 \ln(22 u^{-1} + 9 u^{-0.3}) + 0.4 \sqrt{\pi u}, \quad (38)$$

which is stated to be accurate to $\sim 1.5\%$ over the full range of u .

2.4 Dielectric tensor and polarisation weights

The mode-resolved opacity requires the polarisation weights $|e_{j,\alpha}|^2$ for the two normal modes ($j = 1$: extraordinary, $j = 2$: ordinary). These are computed from the cold-plasma dielectric tensor (`dielectric_tensor.jl`) via the Stix parameters (Ginzburg 1970; Shafranov 1967):

$$S = 1 - \frac{\omega_{pe}^2}{\omega^2 - \omega_{ce}^2} - \frac{\omega_{pi}^2}{\omega^2 - \omega_{cp}^2}, \quad D = \frac{\omega_{ce} \omega_{pe}^2}{\omega(\omega^2 - \omega_{ce}^2)} - \frac{\omega_{cp} \omega_{pi}^2}{\omega(\omega^2 - \omega_{cp}^2)}, \quad P = 1 - \frac{\omega_{pe}^2 + \omega_{pi}^2}{\omega^2}, \quad (39)$$

where $\omega_{pe}^2 = 4\pi n_e e^2 / m_e$ and $\omega_{pi}^2 = 4\pi n_e e^2 / m_p$. The mode parameter q (PC03 Eq. 25) is

$$q = \frac{(P - S) \sin^2 \theta_B}{2 D \cos \theta_B}, \quad (40)$$

where θ_B is the angle between the photon propagation direction and $\mathbf{B}$. The transverse polarisation ratios follow from the quadratic $K^2 + 2qK - 1 = 0$ (Shafranov 1967):

$$K_1 = -q + \sqrt{1 + q^2} \quad (\text{extraordinary}), \quad K_2 = -q - \sqrt{1 + q^2} \quad (\text{ordinary}). \quad (41)$$

Note that $K_1 K_2 = -1$. The longitudinal component is

$$K_{z,j} = -\frac{K_j (S - K_j D) \sin \theta_B \cos \theta_B}{P - (S - K_j D) \sin^2 \theta_B}, \quad (42)$$

from which the cyclic weights are extracted:

$$|e_{j,+1}|^2 = \frac{(K_j + 1)^2}{2\mathcal{N}_j}, \quad |e_{j,-1}|^2 = \frac{(K_j - 1)^2}{2\mathcal{N}_j}, \quad |e_{j,0}|^2 = \frac{K_{z,j}^2}{\mathcal{N}_j}, \quad (43)$$

with normalisation $\mathcal{N}_j = 1 + K_j^2 + K_{z,j}^2$. The weights satisfy $\sum_\alpha |e_{j,\alpha}|^2 = 1$ after a safety renormalisation clamp.

Limiting cases. At $\theta_B = 0$ the modes are purely circularly polarised: mode 1 has all weight on $\alpha = +1$, mode 2 on $\alpha = -1$. At $\theta_B = \pi/2$ mode 1 is purely longitudinal ($\alpha = 0$) and mode 2 is an equal mix of $\alpha = \pm 1$. These limits are hard-coded as special cases to avoid division by zero in Eq. (40).

Design choice. An earlier version of this code used the approximate relation $q(\omega, \theta_B) \approx \bar{q}(\omega) \sin^2 \theta_B / (2 \cos \theta_B)$ (PC03 Eq. 26), which is inaccurate near cyclotron resonances. The current implementation uses the full Stix-parameter formulation of Eq. (40), which is numerically stable at all densities because q is determined primarily by ω_{ce}/ω .

2.5 Normal-mode opacities and the absorption/scattering split

Given the polarisation weights, the opacity for normal mode j at angle θ_B is (PC03 Eq. 27; `magnetic_modes.jl`, function `mode_opacity`):

$$\kappa_j(\theta_B) = \frac{1}{m_H} \sum_{\alpha=-1}^{+1} |e_{j,\alpha}(\theta_B)|^2 \sigma_{\alpha}^{\text{tot}}, \quad (44)$$

where $m_H = m_p + m_e$ and $\sigma_{\alpha}^{\text{tot}}$ is given by Eq. (31).

The code provides a clean *absorption/scattering split* for use by the Feautrier transport solver, which requires them separately:

$$\kappa_j^{\text{abs}}(\theta_B) = \frac{1}{m_H} \sum_{\alpha} |e_{j,\alpha}|^2 (\sigma_{\alpha}^{\text{ff}} + \sigma_{\alpha}^{\text{pp}}), \quad (45)$$

$$\kappa_j^{\text{scat}}(\theta_B) = \frac{1}{m_H} \sum_{\alpha} |e_{j,\alpha}|^2 \sigma_{\alpha}^{s,e}. \quad (46)$$

These are implemented as the functions `mode_absorption` and `mode_scattering` in `magnetic_modes.jl`. By construction, $\kappa_j = \kappa_j^{\text{abs}} + \kappa_j^{\text{scat}}$.

Effective (unpolarised) opacity. For unpolarised radiation at a single angle, the effective opacity is the harmonic mean of the two normal modes:

$$\kappa_{\text{eff}} = \frac{2}{1/\kappa_1 + 1/\kappa_2}. \quad (47)$$

2.6 Angle-averaged opacities and Rosseland means

The diffusion approximation for transport along ($\parallel$) and across ($\perp$) the magnetic field introduces angle-averaged opacities for each mode separately (PC03 Eq. 30; `magnetic_modes.jl`, function `kappa_parallel_perp_mono`):

$$\frac{1}{\kappa_j^{\parallel}} = \frac{3}{4} \int_0^{\pi} \frac{\cos^2 \theta_B}{\kappa_j(\theta_B)} \sin \theta_B \, d\theta_B, \quad \frac{1}{\kappa_j^{\perp}} = \frac{3}{2} \int_0^{\pi} \frac{\sin^3 \theta_B}{\kappa_j(\theta_B)} \, d\theta_B. \quad (48)$$

These integrals are evaluated by the midpoint rule with $N_{\theta} = 40$ uniform panels. After obtaining $\kappa_j^{\parallel}$ and $\kappa_j^{\perp}$ for each mode $j \in \{1, 2\}$, the code forms the harmonic mean over modes:

$$\kappa_{\text{eff}}^{\parallel} = \frac{2}{1/\kappa_1^{\parallel} + 1/\kappa_2^{\parallel}}, \quad \kappa_{\text{eff}}^{\perp} = \frac{2}{1/\kappa_1^{\perp} + 1/\kappa_2^{\perp}}. \quad (49)$$

Importance of mode-separate averaging. A previous version of the code incorrectly combined the two normal modes *before* angle-averaging, which overpredicts the opacity when the two modes differ by orders of magnitude (as they typically do near the cyclotron resonance). The current implementation performs the angle-average on each mode independently, then combines them, following the prescription of PC03 Eq. 30.

2.6.1 Magnetic Rosseland means

The Rosseland mean opacities for transport along and across the field are (`magnetic_modes.jl`, function `rosseland_magnetic`):

$$\frac{1}{K_{\parallel}} = \frac{\int_0^{\infty} \frac{1}{\kappa_{\text{eff}}^{\parallel}(\nu)} \frac{\partial B_{\nu}}{\partial T} d\nu}{\int_0^{\infty} \frac{\partial B_{\nu}}{\partial T} d\nu}, \quad \frac{1}{K_{\perp}} = \frac{\int_0^{\infty} \frac{1}{\kappa_{\text{eff}}^{\perp}(\nu)} \frac{\partial B_{\nu}}{\partial T} d\nu}{\int_0^{\infty} \frac{\partial B_{\nu}}{\partial T} d\nu}. \quad (50)$$

These correspond to columns 13 and 14 of the Potekhin opacity tables (PC03, Table 1).

The frequency grid is constructed by `make_rosseland_frequency_grid`: $N_{\text{base}} = 200$ logarithmically spaced points spanning $[0.01, 200] \times k_B T/h$, augmented with $N_{\text{res}} = 20$ additional points clustered within ± 0.3 dex of each cyclotron resonance (ν_{ce} and ν_{cp}) to resolve the sharp features in the cross-sections near these frequencies.

2.7 Summary of equation cross-references

Table 2 maps every implemented equation to its source in PC03 and to the corresponding Julia function.

Table 2: Cross-reference of opacity equations to Potekhin & Chabrier (2003) and source code.

Report Eq.	PC03 Eq.	Source file / function	Description
(22)	(1)	<code>magnetic_ff / beta_e</code>	Quantisation parameter β_e
(40)	(25)	<code>dielectric_tensor / polarization_weights_full</code>	Mode parameter q
(44)	(27)	<code>magnetic_modes / mode_opacity</code>	Mode opacity $\kappa_j(\theta_B)$
(48)	(30)	<code>magnetic_modes / kappa_parallel_perp_mono</code>	Angle-averaged $\kappa_j^{\parallel, \perp}$
(23)	(33)	<code>magnetic_ff / sigma_scatter_alpha</code>	Magnetic Thomson $\sigma_{\alpha}^{s,e}$
(25)	(37,51)	<code>magnetic_ff / sigma_ff_alpha</code>	Free-free cross-section $\sigma_{\alpha}^{\text{ff}}$
(24)	(39)	<code>magnetic_ff / sigma_ff_alpha (inline)</code>	Radiative damping ν_e^s
(27)	(41)	<code>magnetic_ff / nu_ff_alpha</code>	Free-free collision freq. ν_{α}^{ff}
(32)	(43)	<code>coulomb_magnetic / coulomb_log_classical_safe</code>	Classical Coulomb log $\Lambda_{\text{cl}}^{\text{ff}}$
(33)	(44a)	<code>coulomb_magnetic / coulomb_log_magnetic</code>	Magnetic Coulomb log $\Lambda_{\alpha}^{\text{ff}}$
(34)–(37)	(44b–e)	<code>coulomb_magnetic / _Q_n_alpha</code>	Integrand components
(28)	(46)	<code>magnetic_ff / sigma_pp_alpha</code>	Proton-proton $\sigma_{\alpha}^{\text{pp}}$
(30)	(47)	<code>magnetic_ff / nu_pp</code>	Proton collision freq. ν^{pp}
(38)	(48)	<code>magnetic_ff / coulomb_log_proton</code>	Proton Coulomb log Λ_{pp}
(26)	(52)	<code>magnetic_ff / sigma_ff_alpha (inline)</code>	Combined damping $\tilde{\nu}_{\alpha}$
(29)	(53)	<code>magnetic_ff / sigma_pp_alpha (inline)</code>	Proton damping $\nu_{p,\alpha}$

2.8 Discrepancies and implementation notes

- Proton scattering omitted.** PC03 Eq. (34) gives the proton scattering cross-section $\sigma_{\alpha}^{s,p} = (m_e/m_p)^2 \omega^2 / [(\omega - \alpha\omega_{\text{cp}})^2 + \nu_{p,\alpha}^2] \sigma_T$. This term is not included in `sigma_total_alpha` (Eq. 31). The omission is justified: away from the proton cyclotron resonance, $\sigma_{\alpha}^{s,p} / \sigma_{\alpha}^{s,e} \sim (m_e/m_p)^2 \approx 3 \times 10^{-7}$. At $\omega \approx \omega_{\text{cp}}$ the proton resonance is already captured in $\sigma_{\alpha}^{\text{pp}}$ (Eq. 28), so the missing scattering contribution affects only the scattering-vs-absorption decomposition at that resonance, not the total opacity.
- Traditional vs. improved free-free formula.** PC03 note that the traditional factorisation (their Eq. 45, $\sigma_{\alpha}^{\text{a}} = \sigma_{\alpha}^{\text{ff}} / \sigma_{\alpha}^{s,e}$) is “erroneous” because it misses interference between terms with changing and fixed Landau quantum number n . Our code implements the improved Eq. (51) directly (Eq. 25), which correctly incorporates the proton-cyclotron suppression factor $(\omega - \alpha\omega_{\text{cp}})^2$ in the denominator.

3. ν_α^{ff} **uses** $\hbar\omega$, **not** $h\nu$. The magnetic module defines $u = \hbar\omega/(k_B T)$ in Eq. (27), whereas the non-magnetic module (`hydrogen_ff.jl`) uses $u = h\nu/(k_B T)$. Since $\hbar\omega = h\nu$, these are identical; however, the intermediate code in `nu_ff_alpha` divides by $\hbar\omega$ (not $h\nu$), which is consistent with the convention of PC03 Eq. 41 where frequencies appear as angular frequencies ω .
4. **Coulomb logarithm switching threshold.** The code switches from the classical $\Lambda_{\text{cl}}^{\text{ff}}$ to the magnetic $\Lambda_\alpha^{\text{ff}}$ at $\beta_e = 0.1$. PC03 state that the non-magnetic Gaunt factor is “applicable if $\beta_e < 1$ ” (their text below Eq. 43). The code’s more conservative threshold of $\beta_e = 0.1$ ensures that the magnetic expression is used whenever the field produces even modest Landau quantisation, at the cost of evaluating the more expensive sum+integral for weakly quantising fields.
5. **Integral truncation in $\Lambda_\alpha^{\text{ff}}$.** The semi-infinite integral $\int_0^\infty Q_n^\alpha dy$ in Eq. (33) is truncated at $y = 20$ in the code. For $\beta_e \gg 1$ the integrand decays exponentially for $y \gg 1$ (via the Bessel functions K_0, K_1 of argument x_n), so this upper bound is adequate. For marginally quantising fields ($\beta_e \sim 0.1$ – 1) the decay is slower but the code falls back to $\Lambda_{\text{cl}}^{\text{ff}}$ at $\beta_e < 0.1$, so the truncation error is bounded.
6. **Gaunt factor source.** The non-magnetic Gaunt factor table (`gaunt_ff.jl`) interpolates the Sutherland (1998) numerical table rather than the Hummer (1988) Padé fit used in PC03. For free-free emission by hydrogen at the temperatures of interest ($T \sim 10^{5.3}$ – 10^7 K), the two sources agree to better than 1%, so this difference is inconsequential.

3 Radiative Transfer

This section documents Module 3 of the neutron star visualization pipeline: the self-consistent radiative transfer atmosphere solver. The non-magnetic solver follows the method of Haakonsen et al. (2012) (hereafter H12), while the magnetic extension adopts the two-mode formalism of Suleimanov et al. (2009) (hereafter SPW09). Throughout, we cite the specific equations from these references that each numerical component implements.

3.1 Atmosphere Structure

We model a plane-parallel, fully ionised hydrogen atmosphere in hydrostatic and radiative equilibrium. The assumptions are those common to the three standard unmagnetized codes reviewed in H12 §2: complete ionisation, coherent scattering, no electron conduction, no self-irradiation, and negligible radiation pressure ($T_{\text{eff}} \ll 10^7$ K).

3.1.1 Hydrostatic equilibrium and equation of state

Neglecting radiation pressure, the gas pressure at column depth y (measured inward from the surface in g cm^{-2}) satisfies

$$P(y) = g_s y, \quad (51)$$

where g_s is the surface gravity, taken to be constant across the atmosphere (H12 Eq. 1; SPW09 Eq. 1 in the $g_{\text{rad}} \rightarrow 0$ limit).

For fully ionised hydrogen the ideal-gas equation of state gives

$$P = (n_e + n_p) k_B T = \frac{2\rho}{m_p} k_B T \quad \implies \quad \rho = \frac{m_p P}{2 k_B T}, \quad (52)$$

corresponding to a mean molecular weight $\mu = 0.5$. This is the same relation used in H12 (via the OPAL EOS tables, which reduce to Eq. (52) in the fully ionised, low-density regime) and in SPW09 Eq. 11.

The column depth is discretised into N logarithmically spaced points,

$$y_i = 10^{\log_{10} y_{\min} + (i-1) \Delta \log y}, \quad i = 1, \dots, N, \quad (53)$$

with default bounds $y_{\min} = 10^{-9} \text{ g cm}^{-2}$ and $y_{\max} = 10^2 \text{ g cm}^{-2}$ (extended automatically until the monochromatic optical depth at the highest frequency exceeds 80; cf. H12 §3.1). Frequency is discretised into K log-spaced points spanning $0.05 k_B T_{\text{eff}}/h \leq \nu \leq 120 k_B T_{\text{eff}}/h$, matching the McPHAC grid bounds.

3.1.2 Eddington temperature profile

The initial temperature profile is obtained from the Eddington approximation (H12 Eq. 10),

$$T^4(\tau_R) = \frac{3}{4} T_{\text{eff}}^4 \left(\tau_R + \frac{2}{3} \right), \quad (54)$$

where τ_R is the Rosseland-mean optical depth. The Rosseland mean opacity k_R is computed from the free-free absorption coefficient and the temperature-derivative-weighted frequency integral

$$\frac{1}{k_R} = \frac{\int_0^\infty \frac{1}{k_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}. \quad (55)$$

The surface boundary condition $T(y=0) = 0.265 T_{\text{eff}}$ follows the McPHAC convention (H12 §3.1). Equation (54) is integrated inward by evaluating $\tau_R \approx k_R y$ with a self-consistently updated $k_R(T, \rho)$ at each depth.

3.2 Feautrier Method

3.2.1 Second-order transfer equation

Defining the Feautrier mean intensity at angle μ_j and frequency ν_k as

$$P_\nu(y, \mu) \equiv \frac{1}{2} [I_\nu(y, \mu) + I_\nu(y, -\mu)], \quad (56)$$

the radiative transfer equation can be written in the second-order form (H12 Eq. 2)

$$\mu^2 \frac{d^2 P_\nu}{d\tau_\nu^2} = P_\nu(y, \mu) - S_\nu(y, \mu), \quad (57)$$

where $d\tau_\nu = k_\nu dy$ with $k_\nu = \kappa_\nu + \sigma_{T,r}$ the total (absorption + reduced Thomson scattering) opacity. For coherent, anisotropic dipole Thomson scattering the phase function is (H12 Eq. 16)

$$\frac{d\sigma_{T,r}}{d\mu'}(\mu, \mu_j) = \sigma_{T,r} \frac{3}{16} (3 + 3\mu^2 \mu_j^2 - \mu^2 - \mu_j^2), \quad (58)$$

which replaces the isotropic factor $\sigma_{T,r}/2$ used in earlier codes (Z96, G02, H06). The source function is (H12 Eq. 4)

$$S_\nu(y, \mu) = (1 - \rho_\nu) B_\nu(T) + 2\rho_\nu \sum_{j'=1}^M P_\nu(y, \mu_{j'}) \frac{d\sigma_{T,r}}{d\mu'}(\mu, \mu_{j'}) a_{j'}, \quad (59)$$

where $\rho_\nu = \sigma_{T,r}/k_\nu$ is the scattering albedo (H12 Eq. 6), B_ν is the Planck function, and $a_{j'}$ are the Gauss–Legendre quadrature weights on $[0, 1]$.

3.2.2 Block-tridiagonal system

Discretising Eq. (57) on the column-depth grid following Auer (1976) (H12 §3.3, Eq. 13) gives the block-tridiagonal system

$$\mathbf{A}_i \mathbf{P}_{i-1} + \mathbf{B}_i \mathbf{P}_i + \mathbf{C}_i \mathbf{P}_{i+1} = \mathbf{Q}_i, \quad i = 1, \dots, N_{\text{eff}}, \quad (60)$$

where $\mathbf{P}_i$ is the M -vector of Feautrier means at depth y_i and the $M \times M$ matrices encode the finite-difference stencil. For interior points $2 \leq i < N_{\text{eff}}$, defining the optical depth increments

$$\Delta\tau_i = \frac{1}{2} (k_\nu(y_i) + k_\nu(y_{i-1})) (y_i - y_{i-1}) \quad (61)$$

and $\overline{\Delta\tau}_i = \frac{1}{2} (\Delta\tau_i + \Delta\tau_{i+1})$, the matrix elements are (diagonal in the angle index j):

$$(A_i)_{jj} = -\frac{\mu_j^2}{\Delta\tau_i \overline{\Delta\tau}_i}, \quad (62)$$

$$(C_i)_{jj} = -\frac{\mu_j^2}{\Delta\tau_{i+1} \overline{\Delta\tau}_i}, \quad (63)$$

$$(B_i)_{jj} = -(A_i)_{jj} - (C_i)_{jj} + 1, \quad (64)$$

with the scattering redistribution contribution

$$(B_i)_{j,j'} \leftarrow (B_i)_{j,j'} - 2\rho_\nu \frac{d\sigma_{T,r}}{d\mu'}(\mu_j, \mu_{j'}) a_{j'}, \quad (65)$$

and the thermal source vector $Q_i = (1 - \rho_\nu) B_\nu(T_i) \mathbf{1}$ (H12 Eqs. 14–15).

3.2.3 Boundary conditions

Surface ($i = 1$). No incident radiation, $I_\nu^-(0, \mu) = 0$, gives the Auer boundary (H12 §3.3):

$$(1 + \delta_j^{-1}) P_1^{(j)} - \delta_j^{-1} P_2^{(j)} = 0, \quad \text{where } \delta_j = \Delta\tau_2/\mu_j. \quad (66)$$

The surface carries no thermal source ($Q_1 = 0$); the emergent intensity is recovered as $I_\nu(0, \mu_j) = 2 P_\nu(y_1, \mu_j)$.

Bottom ($i = N_{\text{eff}}$). The diffusion (LTE) limit sets $P_\nu = B_\nu(T_{N_{\text{eff}}})$ for all angles, ensuring isotropic specific intensity at great depth (H12 §3.3). The effective depth N_{eff} is chosen so that $\tau_{\nu, \text{max}} = 80$ at each frequency.

3.2.4 Forward elimination and back-substitution

The system (60) is solved by Gaussian elimination on the block matrices. Define the modified quantities recursively:

$$\tilde{\mathbf{B}}_1 = \mathbf{B}_1, \quad \tilde{\mathbf{Q}}_1 = \mathbf{Q}_1, \quad (67)$$

$$\tilde{\mathbf{B}}_i = \mathbf{B}_i - \mathbf{A}_i \tilde{\mathbf{B}}_{i-1}^{-1} \mathbf{C}_{i-1}, \quad (68)$$

$$\tilde{\mathbf{Q}}_i = \mathbf{Q}_i - \mathbf{A}_i \tilde{\mathbf{B}}_{i-1}^{-1} \tilde{\mathbf{Q}}_{i-1}. \quad (69)$$

Back-substitution proceeds as

$$\mathbf{P}_{N_{\text{eff}}} = \tilde{\mathbf{B}}_{N_{\text{eff}}}^{-1} \tilde{\mathbf{Q}}_{N_{\text{eff}}}, \quad \mathbf{P}_i = \tilde{\mathbf{B}}_i^{-1} (\tilde{\mathbf{Q}}_i - \mathbf{C}_i \mathbf{P}_{i+1}). \quad (70)$$

A positivity clamp $P_{i,j} \leftarrow \max(P_{i,j}, 0)$ is applied after back-substitution to ensure physical specific intensities. Gauss–Legendre quadrature over the outward hemisphere then yields the moments

$$J_\nu(y_i) = \sum_{j=1}^M a_j P_\nu(y_i, \mu_j), \quad f_\nu = \frac{\sum_j \mu_j^2 a_j P_j}{J_\nu}, \quad h_\nu = \frac{\sum_j \mu_j a_j P_j}{J_\nu}, \quad (71)$$

where f_ν and h_ν are the Eddington factors (H12 Eqs. 19–20, 22).

3.3 Frequency-Adaptive Depth Grid

At a given frequency, only column depths with $\tau_\nu \lesssim 80$ contribute to the emergent flux. Different frequencies become optically thick at vastly different physical depths because $\kappa_{\text{ff}} \propto \nu^{-3}$; a single common grid therefore wastes resolution at some frequencies while under-resolving others.

Following the McPHAC approach (H12 §3.3), an adaptive variant constructs a per-frequency depth grid. For each frequency ν_k :

1. Determine $y_{\text{max},k}$ such that $\tau_{\nu_k}(y_{\text{max},k}) = 80$.
2. Create N_ν logarithmically spaced points $\{y_i^{(\nu)}\}$ from y_{min} to $y_{\text{max},k}$.
3. Interpolate $T(y)$, $k_\nu(y)$, and $\rho_\nu(y)$ from the common grid onto $\{y_i^{(\nu)}\}$ using log-linear interpolation (logarithmic in y , linear in the field quantity).
4. Solve the Feautrier system (60) on the per-frequency grid.
5. Interpolate the resulting J_ν , f_ν , h_ν back to the common grid.

This concentrates depth resolution near each frequency’s photosphere and improves spectral accuracy from $\sim 7\%$ (common grid) to $< 1\%$ (adaptive grid), at the cost of K independent interpolation steps per iteration.

For depths beyond $y_{\text{max},k}$ the diffusion limit $J_\nu = B_\nu(T)$, $f_\nu = 1/3$, $h_\nu = 0$ is imposed.

3.4 Rybicki Temperature Correction

The Feautrier solution yields J_ν and f_ν for a *fixed* temperature profile. To drive the atmosphere toward radiative equilibrium (constant bolometric flux through all depths), we apply the Rybicki temperature correction method (Rybicki, 1971), following the detailed discretisation in H12 Appendix A.

3.4.1 Linearised moment equation

Integrating the second-order transfer equation over angle (H12 Eq. A1) and using the Eddington factor closure yields (H12 Eq. A2)

$$\frac{1}{k_\nu} \frac{d}{dy} \left(\frac{1}{k_\nu} \frac{d}{dy} (f_\nu J_\nu) \right) = (J_\nu - B_\nu) (1 - \rho_\nu). \quad (72)$$

Radiative equilibrium requires $\int_0^\infty (J_\nu - B_\nu) \kappa_\nu d\nu = 0$ at every depth (H12 Eq. A3). Writing $T = T_0 + \Delta T$ and linearising $B_\nu(T) \approx B_\nu(T_0) + (dB_\nu/dT) \Delta T$ gives the temperature correction (H12 Eq. A5):

$$\Delta T(y) = \frac{\int_0^\infty (J_\nu - B_\nu(T_0)) \kappa_\nu d\nu}{\int_0^\infty \frac{dB_\nu}{dT} \kappa_\nu d\nu}. \quad (73)$$

Substituting this back into Eq. (72) and eliminating ΔT produces a coupled system for J_ν at all depths and frequencies simultaneously (H12 Eq. A6).

3.4.2 Tridiagonal system per frequency

Discretising Eq. (72) on the column-depth grid and substituting the linearised ΔT produces, for each frequency k , a tridiagonal system (H12 Eq. A18):

$$\mathbf{T}_k \mathbf{J}_k + \mathbf{U}_k \bar{\mathbf{J}} = \mathbf{K}_k, \quad (74)$$

where $\mathbf{J}_k = (J_{k,1}, \dots, J_{k,N})^T$ is the mean intensity at frequency k , $\bar{\mathbf{J}}$ is the frequency-integrated weighted mean (defined below), and the tridiagonal matrix elements are (H12 Eqs. A19–A21, with the notation of Eqs. A12–A14):

$$(\mathbf{T}_k)_{i,i-1} = -\frac{8 f_k(y_{i-1})}{\pi_i \Delta_1}, \quad (75)$$

$$(\mathbf{T}_k)_{i,i+1} = -\frac{8 f_k(y_{i+1})}{\pi_i \Delta_2}, \quad (76)$$

$$(\mathbf{T}_k)_{i,i} = \left[\frac{1 - \rho_k}{f_k(y_i)} + \frac{8}{\pi_i \Delta_1} + \frac{8}{\pi_i \Delta_2} \right] f_k(y_i), \quad (77)$$

where $\Delta_1 = [k_\nu(y_i) + k_\nu(y_{i-1})] (y_i - y_{i-1})$, $\Delta_2 = [k_\nu(y_{i+1}) + k_\nu(y_i)] (y_{i+1} - y_i)$, and $\pi_i = \Delta_1 + \Delta_2$ (H12 Eqs. A12–A14). The coupling vector and source are (H12 Eqs. A22–A23):

$$(U_k)_i = -\frac{dB_k}{dT} (1 - \rho_k), \quad (78)$$

$$(K_k)_i = \left(B_k(T_0) - \frac{dB_k}{dT} \bar{B}_i \right) (1 - \rho_k), \quad (79)$$

where

$$\bar{B}_i \equiv \frac{\sum_{k'=1}^K B_{k'}(T_0) \kappa_{k'} b_{k'}}{\sum_{k'=1}^K \frac{dB_{k'}}{dT} \kappa_{k'} b_{k'}} \quad (80)$$

and b_k are the trapezoidal frequency quadrature weights.

3.4.3 Global $\bar{J}$ system

The key insight of the Rybicki method is to define the global weighted mean intensity (H12 Eq. A17)

$$\bar{J}_i = \frac{\sum_{k=1}^K J_{k,i} \kappa_k b_k}{\sum_{k=1}^K \frac{dB_k}{dT} \kappa_k b_k}, \quad (81)$$

so that $\Delta T_i = \bar{J}_i - \bar{B}_i$ (from Eq. 73). Inverting the tridiagonal $\mathbf{T}_k$ for each frequency and substituting into Eq. (74) yields a dense $N \times N$ system (H12 Eq. A24):

$$\left(\mathbf{I} + \sum_{k=1}^K \mathbf{V}_k \mathbf{T}_k^{-1} \text{diag}(\mathbf{U}_k) \right) \bar{\mathbf{J}} = \sum_{k=1}^K \mathbf{V}_k \mathbf{T}_k^{-1} \mathbf{K}_k, \quad (82)$$

where $\mathbf{V}_k$ is the diagonal matrix with entries (H12 Eq. A25)

$$(V_k)_{i,i'} = \frac{\kappa_k b_k}{\sum_{k'} \frac{dB_{k'}}{dT} \kappa_{k'} b_{k'}} \delta_{i,i'}. \quad (83)$$

The $\mathbf{T}_k^{-1}$ factors are never formed explicitly. Instead, for each frequency k , the Thomas algorithm (LU factorisation of the tridiagonal) is used to solve $\mathbf{T}_k \mathbf{x} = \mathbf{K}_k$ and $\mathbf{T}_k \mathbf{z}_j = \mathbf{e}_j$ for each column j where $U_{k,j} \neq 0$. This gives the contributions to the right-hand side and the $\mathbf{W}$ matrix,

$$W_{ij} = \sum_k (V_k)_i (U_k)_j (\mathbf{T}_k^{-1})_{ij}, \quad (84)$$

after which the $N \times N$ dense system $(\mathbf{W} + \mathbf{I}) \bar{\mathbf{J}} = \mathbf{r}$ is solved by direct factorisation.

3.4.4 Boundary conditions for the Rybicki system

At the surface ($i = 1$), the flux boundary condition (H12 Eqs. A26, A28–A30) is discretised as

$$\left(h_k + \frac{f_k(y_1)}{\Delta_{\text{surf}}} \right) J_{k,1} - \frac{f_k(y_2)}{\Delta_{\text{surf}}} J_{k,2} = 0, \quad (85)$$

where $\Delta_{\text{surf}} = \frac{1}{2}[k_\nu(y_1) + k_\nu(y_2)](y_2 - y_1)$ and $h_k = h_\nu(y_1)$ is the flux Eddington factor from the Feautrier solution. Both $U_{k,1}$ and $K_{k,1}$ vanish at the surface; the surface temperature correction enters solely through the global $\bar{\mathbf{J}}$ coupling.

At the bottom ($i = N$), the LTE condition $J_\nu = B_\nu$ sets $(\mathbf{T}_k)_{N,N} = 1$, $K_{k,N} = B_k(T_N)$, $U_{k,N} = 0$ (H12 Eqs. A31–A33).

3.5 Two-Mode Magnetic Extension

In a magnetised atmosphere with field strength B and inclination θ_B to the surface normal, radiation propagates in two normal modes—the extraordinary (X, $j=1$) and ordinary (O, $j=2$)—with distinct, mode-dependent opacities $\kappa_\nu^{(j)}$, $\sigma_\nu^{(j)}$ (SPW09 §2, Eqs. 5–8). The implementation follows SPW09 for the opacity framework and extends the H12 iteration scheme to two modes.

3.5.1 Per-mode transfer equation

Each mode satisfies its own transfer equation (SPW09 Eq. 5):

$$\mu \frac{dI_\nu^{(j)}}{d\tau_\nu^{(j)}} = I_\nu^{(j)} - S_\nu^{(j)}, \quad (86)$$

where $d\tau_\nu^{(j)} = (k_\nu^{(j)} + \sigma_\nu^{(j)}) dm$ and the per-mode source function is (SPW09 Eq. 6)

$$S_\nu^{(j)} = \frac{k_\nu^{(j)}}{k_\nu^{(j)} + \sigma_\nu^{(j)}} \frac{B_\nu}{2} + \frac{\sigma_\nu^{(j)}}{k_\nu^{(j)} + \sigma_\nu^{(j)}} (\text{scattering integral}). \quad (87)$$

The factor of 1/2 in the thermal term arises because unpolarised blackbody emission is split equally between the two modes. This is the key distinction from the non-magnetic case: each mode sees $B_\nu/2$ rather than B_ν as its thermal source (SPW09 Eq. 6; cf. SPW09 Eq. 15 for the bottom boundary $I_\nu^{(j)}(\mu > 0) = B_\nu/2$).

The Feautrier equation for mode j at interior points takes the same block-tridiagonal form as Eq. (60), with the replacements $k_\nu \rightarrow k_\nu^{(j)} + \sigma_\nu^{(j)}$, $\rho_\nu \rightarrow \sigma_\nu^{(j)} / (k_\nu^{(j)} + \sigma_\nu^{(j)})$, and $B_\nu \rightarrow B_\nu/2$. The two modes are solved independently at each iteration: the coupling between them enters only through the shared temperature profile and its correction.

3.5.2 Two-mode Rybicki correction

Radiative equilibrium in the two-mode case requires (SPW09 Eq. 9, simplified to the plane-parallel case with mode-averaged opacities)

$$\sum_{j=1}^2 \int_0^\infty \kappa_\nu^{(j)} \left(J_\nu^{(j)} - \frac{B_\nu}{2} \right) d\nu = 0 \quad (88)$$

at every depth. The Rybicki system (74) is extended by summing over both modes: the global quantities become

$$\bar{B}_i = \frac{\sum_{j=1}^2 \sum_{k=1}^K \frac{1}{2} B_k(T_0) \kappa_k^{(j)} b_k}{\sum_{j=1}^2 \sum_{k=1}^K \frac{1}{2} \frac{dB_k}{dT} \kappa_k^{(j)} b_k}, \quad (89)$$

$$(V_{k,j})_i = \frac{\kappa_k^{(j)} b_k}{\sum_{j'} \sum_{k'} \frac{1}{2} \frac{dB_{k'}}{dT} \kappa_{k'}^{(j')} b_{k'}}, \quad (90)$$

and per-mode tridiagonal systems $\mathbf{T}_k^{(j)}$ are built with mode-specific opacities and Eddington factors $f_k^{(j)}$, $h_k^{(j)}$. The per-mode coupling vector and source are

$$(U_k^{(j)})_i = -\frac{1}{2} \frac{dB_k}{dT} (1 - \rho_k^{(j)}), \quad (91)$$

$$(K_k^{(j)})_i = \left(\frac{1}{2} B_k - \frac{1}{2} \frac{dB_k}{dT} \bar{B}_i \right) (1 - \rho_k^{(j)}), \quad (92)$$

where $\bar{B}_i$ is the *global* average from Eq. (89) (not a per-mode quantity). The dense system (82) accumulates contributions from both modes:

$$W_{i,i'} = \sum_{j=1}^2 \sum_{k=1}^K (V_{k,j})_i (U_k^{(j)})_{i'} [(\mathbf{T}_k^{(j)})^{-1}]_{i,i'}, \quad (93)$$

and the system $(\mathbf{W} + \mathbf{I}) \bar{\mathbf{J}} = \mathbf{r}$ is solved identically to the non-magnetic case.

3.5.3 Uniform flux correction

In addition to the Rybicki correction, a uniform flux correction is applied when the bolometric flux error exceeds 2%. The total emergent flux is

$$F_{\text{bol}} = 2\pi \sum_{j=1}^2 \sum_k \Delta\nu_k \sum_{l=1}^M \mu_l a_l 2 P_{\nu_k}^{(j)}(y_1, \mu_l), \quad (94)$$

and the correction $\delta T_i = -\alpha T_i (F_{\text{bol}}/\sigma T_{\text{eff}}^4 - 1)/4$ with $\alpha = 0.1$ is added to the Rybicki ΔT before application (cf. the Avrett–Krook surface correction of SPW09 §2). This stabilises convergence when the Rybicki correction alone produces flux drift due to mode-asymmetric opacity structure.

For $B = 0$, both modes have identical opacities and $B_\nu/2 + B_\nu/2 = B_\nu$, recovering the non-magnetic single-mode result.

3.6 Convergence Criteria and Iteration Scheme

The full iteration scheme proceeds as follows.

1. **Initialise.** Build the column structure (Eq. 51–52), Eddington T -profile (Eq. 54), and compute opacities at all (y_i, ν_k) grid points.
2. **Feautrier solve.** Solve the block-tridiagonal system (60) at every frequency (and, for the magnetic case, for each mode independently). Extract J_ν , f_ν , h_ν via Eq. (71).
3. **Temperature correction.** Compute the Rybicki ΔT by solving the dense system (82). For the magnetic case, accumulate over both modes via Eq. (93) and optionally add the uniform flux correction.
4. **Damping.** If $\max_i |\Delta T_i/T_i| > 0.3$, uniformly scale ΔT so that the maximum fractional correction is 30%. This prevents divergent oscillations in early iterations.
5. **Apply and update.** Set $T_i \leftarrow \max(T_i + \Delta T_i, 0.1 T_{\text{eff}})$ (floor prevents unphysical cooling). Recompute ρ , κ_ν , k_ν , ρ_ν , and τ_ν from the updated T -profile.
6. **Check convergence.** The iteration converges when

$$\max_i \left| \frac{\Delta T_i}{T_i} \right| < \epsilon, \quad (95)$$

with default tolerance $\epsilon = 10^{-4}$ (matching H12 §3, §4.3). For the magnetic solver, an additional flux criterion $|F_{\text{bol}}/\sigma T_{\text{eff}}^4 - 1| < 0.05$ is required, with a relaxed temperature tolerance of 10ϵ accepted if the flux error is below 3%.

7. **Iterate.** Repeat from step 2 until convergence or a maximum iteration count (default 30) is reached.
8. **Final spectrum.** After convergence, perform one last Feautrier solve and extract the emergent specific intensity $I_\nu(\mu_j) = 2 P_\nu(y_1, \mu_j)$.

At the default discretisation ($K = 50$, $M = 10$, $N = 100$), convergence to $\epsilon = 10^{-4}$ is typically achieved within 5–8 iterations for $T_{\text{eff}} \sim 10^{5.5}–10^{6.5}$ K, consistent with the McPHAC results (H12 Fig. 6). The magnetic solver with $N = 200$ depth points and $M = 8$ angles generally requires 8–15 iterations due to the stronger opacity gradients introduced by the mode-dependent cross-sections.

4 Surface Model and Dipole Geometry

We model the neutron star surface following the framework of Greenstein & Hartke (1983), in which anisotropic heat conduction through the magnetised crust produces a latitude-dependent effective temperature. The magnetic field is taken to be a centred dipole, tilted by an obliquity angle α from the stellar spin axis.

4.1 Magnetic colatitude

At a geographic position $(\theta_{\text{geo}}, \varphi_{\text{geo}})$ on the stellar surface, the magnetic colatitude θ_B with respect to a dipole axis oriented along $\hat{\mathbf{m}} = (\sin \alpha, 0, \cos \alpha)$ is

$$\cos \theta_B = \cos \theta_{\text{geo}} \cos \alpha + \sin \theta_{\text{geo}} \cos \varphi_{\text{geo}} \sin \alpha. \quad (96)$$

This is simply the dot product of the outward surface normal with the magnetic axis unit vector, implemented as `magnetic_colatitude()` in `src/surface/dipole.jl`.

4.2 Greenstein–Hartke temperature map

Greenstein & Hartke (1983) show that the anisotropy of thermal conductivity ($K_{\parallel} \gg K_{\perp}$) along versus across the magnetic field lines channels heat preferentially toward the magnetic poles. Their Eq. (14) gives the general relation

$$T^4 = T_{\text{pole}}^4 \left(\sin^2 \Phi + \frac{K_{\perp}}{K_{\parallel}} \cos^2 \Phi \right), \quad (97)$$

where Φ is the angle between the magnetic field and the vertical, related to the magnetic colatitude β by $\tan \Phi = 2 \cot \beta$ (their Eq. 15). In the limit $K_{\perp}/K_{\parallel} \rightarrow 0$ the equatorial temperature vanishes; for $K_{\perp} = K_{\parallel}$ the surface is isothermal.

Our implementation adopts the simplified parametric form used widely in the literature:

$$T(\theta_B) = T_{\text{eq}} + (T_{\text{pole}} - T_{\text{eq}}) \cos^2 \theta_B, \quad (98)$$

which recovers $T = T_{\text{pole}}$ at the magnetic poles ($\theta_B = 0$) and $T = T_{\text{eq}}$ at the magnetic equator ($\theta_B = \pi/2$). This corresponds to the leading-order behaviour of Eq. (97) when T_{eq} absorbs the residual equatorial flux, and is cited as Eq. (1) of Greenstein & Hartke (1983) in the code documentation. The function `surface_temperature( $\theta_B$ ,  $T_{\text{pole}}$ ,  $T_{\text{eq}}$ )` enforces $T_{\text{pole}} \geq T_{\text{eq}} > 0$.

4.3 Dipole magnetic field strength

For a centred magnetic dipole, the surface field magnitude at magnetic colatitude θ_B is the standard result

$$|\mathbf{B}|(\theta_B) = \frac{B_{\text{pole}}}{2} \sqrt{1 + 3 \cos^2 \theta_B}, \quad (99)$$

giving $|\mathbf{B}| = B_{\text{pole}}$ at the pole and $|\mathbf{B}| = B_{\text{pole}}/2$ at the equator (Greenstein & Hartke, 1983). Both $T(\theta_B)$ and $B(\theta_B)$ serve as inputs to the atmosphere grid lookup described in Section 7.

5 Schwarzschild Geodesic Ray Tracing

Photons propagating outward from the neutron star surface travel along null geodesics of the Schwarzschild metric. We follow the analytic treatment of Beloborodov (2002) and cross-reference with the exact elliptic-integral formulation of Pechenick et al. (1983).

5.1 The cosine relation

Let a photon be emitted at the stellar surface $r = R$ at an angle α to the radial direction. Observed at infinity, this photon escapes at a polar angle ψ from the sub-observer point. Beloborodov (2002), Eq. (1), gives the remarkably accurate approximate relation

$$1 - \cos \alpha = (1 - \cos \psi) \left(1 - \frac{r_g}{R}\right), \quad (100)$$

where $r_g = 2GM/c^2$ is the Schwarzschild radius and $u \equiv r_g/R$ is the compactness parameter. This approximation is valid for $R > 2r_g$ (i.e. $u < 1$) with fractional error $< 1\%$ for the typical neutron star compactness $R \approx 3r_g$ (Beloborodov, 2002, Fig. 2).

Inverting Eq. (100) yields

$$\cos \alpha = 1 - (1 - \cos \psi)(1 - u), \quad (101)$$

$$\cos \psi = 1 - \frac{1 - \cos \alpha}{1 - u}, \quad (102)$$

both implemented in `src/geodesics/schwarzschild.jl`.

5.2 Impact parameter

The impact parameter b is related to the emission angle by (Beloborodov, 2002, Eq. 10)

$$\sin \alpha = \frac{b}{R} \sqrt{1 - \frac{r_g}{R}}, \quad (103)$$

so that

$$b = \frac{R \sin \alpha}{\sqrt{1 - u}}. \quad (104)$$

At the limb of the apparent disc the photon departs tangentially ($\cos \alpha = 0$, $\sin \alpha = 1$), giving the critical impact parameter

$$b_{\text{crit}} = \frac{R}{\sqrt{1 - u}}. \quad (105)$$

This agrees with Eq. (2.10) of Pechenick et al. (1983), $b_{\text{max}} = R(1 - 2M/R)^{-1/2}$, noting the equivalence $2M/R \equiv r_g/R = u$ in their geometrised units ($G = c = 1$).

5.3 Exact bending-angle integral

The exact bending angle is given by an elliptic integral over the photon orbit (Pechenick et al., 1983, Eq. 2.12):

$$\Delta\phi = \int_0^{M/R} [u_b^2 - (1 - 2u)u^2]^{-1/2} du, \quad (106)$$

where $u_b \equiv M/b$ and the integration variable $u \equiv M/r$. Our code uses the Beloborodov approximation (Eq. 100) rather than evaluating this integral numerically, which introduces at most $\sim 1\%$ error for $R \gtrsim 2r_g$ and avoids the need for special-function libraries.

5.4 Visible fraction of the stellar surface

Gravitational light bending allows a distant observer to see more than the Euclidean hemisphere. The fraction of the surface visible is (Beloborodov, 2002, Sect. 3)

$$f_{\text{vis}} = \frac{S_v}{4\pi R^2} = \frac{1}{2(1 - u)}. \quad (107)$$

In flat space ($u \rightarrow 0$), $f_{\text{vis}} \rightarrow 1/2$ as expected. For a canonical neutron star with $R = 3 r_g$ ($u = 2/3$), $f_{\text{vis}} = 3/2 \cdot 1/2 = 3/4$: three-quarters of the surface is visible simultaneously. The corresponding result in Pechenick et al. (1983) follows from $(\Delta\phi)_{\text{max}} = 180^\circ$ when $R/2M = 1.76$ (their p. 848), at which point the entire surface becomes visible.

5.5 Image construction

The `trace_image()` routine constructs an $N \times N$ pixel grid in the observer's image plane with Cartesian coordinates (x, y) spanning $[-b_{\text{max}}, b_{\text{max}}]$ (where $b_{\text{max}} = 1.05 b_{\text{crit}}$). For each pixel:

1. Compute the impact parameter $b = \sqrt{x^2 + y^2}$ and image azimuth $\varphi_{\text{img}} = \arctan(y/x)$.
2. If $b > b_{\text{crit}}$, the ray misses the star.
3. Otherwise, obtain $\sin \alpha = (b/R)\sqrt{1-u}$ from Eq. (103), then $\cos \alpha$ and $\cos \psi$ from Eqs. (101)–(102).
4. Map the sky position $(\psi, \varphi_{\text{img}})$ to surface coordinates (θ_s, φ_s) via a spherical rotation accounting for the observer inclination i :

$$\begin{aligned} x_s &= \sin i \cos \psi + \cos i \sin \psi \cos \varphi_{\text{img}}, \\ y_s &= \sin \psi \sin \varphi_{\text{img}}, \\ z_s &= \cos i \cos \psi - \sin i \sin \psi \cos \varphi_{\text{img}}, \end{aligned} \tag{108}$$

with $\theta_s = \arccos z_s$ and $\varphi_s = \arctan(y_s, x_s)$.

5. Record the gravitational redshift $1+z = (1-u)^{-1/2}$, which is constant over the surface for the Schwarzschild metric.

Each ray result stores $\{b, \psi, \theta_s, \varphi_s, \cos \alpha, 1+z\}$.

6 Relativistic Intensity Transform

The specific intensity I_ν transforms under a gravitational redshift according to the Lorentz-invariance of I_ν/ν^3 (see e.g. Beloborodov, 2002, Sect. 3):

$$\frac{I_\nu^{\text{obs}}}{\nu_{\text{obs}}^3} = \frac{I_{\nu'}^{\text{em}}}{\nu'^3}, \tag{109}$$

where $\nu' = (1+z)\nu_{\text{obs}}$ is the emitted frequency. Rearranging,

$$I_\nu^{\text{obs}}(\nu_{\text{obs}}) = \frac{I_{\nu'}^{\text{em}}(\nu' = (1+z)\nu_{\text{obs}})}{(1+z)^3}. \tag{110}$$

This is the central equation applied per-pixel in the rendering pipeline. The observed flux element from a surface patch dS at angle α to the observer is (Beloborodov, 2002, Eq. 4)

$$dF = \left(1 - \frac{r_g}{R}\right)^2 I_0(\alpha) \cos \alpha \frac{dS}{D^2}, \tag{111}$$

where the $(1 - r_g/R)^2$ factor combines the $(1+z)^{-3}$ intensity suppression with the $(1+z)^{+1}$ boost from the aberration of solid angles. In our pixel-based rendering, each pixel subtends a known solid angle on the sky, so we work directly with the specific intensity (Eq. 110) rather than integrating fluxes.

7 Spectral Image Cube and Atmosphere Grid Lookup

7.1 Pre-computed atmosphere grid

To avoid re-solving the radiative transfer equation at every surface element, we pre-compute atmosphere models on a two-dimensional grid in (T_{eff}, B) and store the emergent specific intensity $I_\nu(\nu, \cos \theta_e)$ for each grid point (`src/pipeline/atmosphere_grid.jl`). The `AtmosphereSpectrumGrid` structure contains:

- A common frequency grid $\{\nu_k\}_{k=1}^K$ and angle-cosine grid $\{\mu_j\}_{j=1}^M$;
- For each (T_i, B_l) : a $K \times M$ matrix of emergent intensities. For magnetic models ($B > 0$), the extraordinary (X) and ordinary (O) mode intensities are summed: $I_{\text{total}} = I^{(X)} + I^{(O)}$.

7.2 Grid interpolation

At runtime, the lookup function `lookup_spectrum( $T_{\text{eff}}, B, \nu_{\text{obs}}, \cos \theta_e$ )` performs trilinear interpolation:

1. *Bilinear in (T_{eff}, B)* : bracket the query values in the pre-computed grids and compute weights w_T, w_B for the four surrounding models.
2. *Linear in angle*: for each of the four corner models, interpolate the $K \times M$ intensity table in the μ -direction at $\mu = \cos \theta_e$.
3. *Nearest-neighbour in frequency*: for each requested ν_{obs} , the nearest grid frequency ν_k is used.

The interpolated intensity at observed frequency ν_{obs} is thus

$$I_\nu^{\text{em}}(\nu', \cos \theta_e) \approx \sum_{a,b \in \{0,1\}} w_T^{(a)} w_B^{(b)} \left[(1 - f_\mu) I_{k,j}^{(a,b)} + f_\mu I_{k,j+1}^{(a,b)} \right], \quad (112)$$

where f_μ is the fractional position within the angle bracket and k is the nearest frequency index.

7.3 Per-pixel spectrum computation

The `render_spectral_cube()` function (`src/pipeline/render.jl`) assembles the full spectral image cube $I(x, y, \nu)$ via the following per-pixel procedure:

1. **Ray trace** $\rightarrow$ surface coordinates (θ_s, φ_s) , emission angle $\cos \alpha$, redshift $1+z$.
2. **Surface model** $\rightarrow$ magnetic colatitude θ_B (Eq. 96), local temperature $T(\theta_B)$ (Eq. 98), local field $B(\theta_B)$ (Eq. 99).
3. **Atmosphere lookup** $\rightarrow$ emitted spectrum $I_{\nu'}^{\text{em}}(\nu_k, \cos \alpha)$ at surface-frame frequencies $\nu_k = (1+z) \nu_{\text{obs},k}$ via Eq. (112).
4. **Redshift** $\rightarrow$ observed intensity per Eq. (110):

$$I_{\nu_k}^{\text{obs}} = \frac{I_{\nu'}^{\text{em}}(\nu_k)}{(1+z)^3}. \quad (113)$$

The result is a data cube `SpectralImageCube`(N_x, N_y, N_ν) storing $I(x, y, \nu)$ in units of $\text{erg s}^{-1} \text{cm}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$, together with per-pixel metadata (hit fraction, local T, B , and $1+z$).

8 CIE 1931 Colorimetry and sRGB Rendering

The spectral image cube must be converted to a displayable colour image. We implement the standard colorimetric pipeline in `src/colorimetry/cie_srgb.jl`.

8.1 CIE 1931 XYZ tristimulus values

The CIE 1931 2° colour matching functions (CMFs) $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ are loaded from the CVRL dataset (360–830 nm at 1 nm resolution). The tristimulus values for an observed spectrum $I_\nu(\nu)$ are obtained by transforming to wavelength space and integrating against the CMFs:

$$X = \int I_\lambda(\lambda) \bar{x}(\lambda) d\lambda, \quad Y = \int I_\lambda(\lambda) \bar{y}(\lambda) d\lambda, \quad Z = \int I_\lambda(\lambda) \bar{z}(\lambda) d\lambda, \quad (114)$$

where the change of variables from frequency to wavelength gives

$$I_\lambda = I_\nu \frac{c}{\lambda^2}, \quad (115)$$

using $|d\nu| = (c/\lambda^2) |d\lambda|$. The integrals are evaluated by trapezoidal quadrature over the CMF wavelength grid, with I_ν interpolated in log-log space (binary search plus linear interpolation of $\ln I_\nu$ vs. $\ln \nu$) to handle the large dynamic range of astrophysical spectra.

8.2 XYZ to linear sRGB

The CIE XYZ tristimulus values are converted to linear sRGB using the IEC 61966-2-1:1999 matrix for the D65 illuminant:

$$\begin{pmatrix} R_{\text{lin}} \\ G_{\text{lin}} \\ B_{\text{lin}} \end{pmatrix} = \begin{pmatrix} 3.2406 & -1.5372 & -0.4986 \\ -0.9689 & 1.8758 & 0.0415 \\ 0.0557 & -0.2040 & 1.0570 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \quad (116)$$

Negative values (out-of-gamut colours) are clamped to zero.

8.3 Tone mapping

Neutron star spectra produce luminance values spanning many orders of magnitude. We apply the global Reinhard tone mapping operator (Reinhard et al., 2002), which compresses the luminance channel Y via

$$L_{\text{scaled}} = \frac{a \cdot Y}{\bar{Y}}, \quad Y_{\text{display}} = \frac{L_{\text{scaled}}}{1 + L_{\text{scaled}}}, \quad (117)$$

where $\bar{Y}$ is the geometric mean luminance of all illuminated pixels and $a = 0.18$ is the key value (scene middle grey). The chromaticity is preserved by scaling all three XYZ channels by Y_{display}/Y before the matrix conversion (Eq. 116).

8.4 sRGB gamma correction

The final step applies the piecewise sRGB transfer function (IEC 61966-2-1):

$$C_{\text{sRGB}} = \begin{cases} 12.92 C_{\text{lin}}, & C_{\text{lin}} \leq 0.0031308, \\ 1.055 C_{\text{lin}}^{1/2.4} - 0.055, & C_{\text{lin}} > 0.0031308, \end{cases} \quad (118)$$

applied independently to each of R , G , B . The result is quantised to 8-bit integers $\in [0, 255]$ and written as a PPM image.

8.5 False-colour X-ray rendering

In addition to the true-colour rendering, the pipeline produces a false-colour X-ray image by integrating the spectral cube over three energy bands and mapping them to RGB channels:

$$\text{Red : 0.1--0.5 keV , \quad Green : 0.5--2.0 keV , \quad Blue : 2.0--10.0 keV .} \quad (119)$$

Each channel is independently normalised to $[0, 255]$, providing a visual diagnostic of spectral hardness variations across the stellar surface.

Summary of Discrepancies Found

During this code review, the following discrepancies between the source code and published references were identified:

1. **Surface gravity comment** (`tov.jl:184`): The comment says “surface gravity (red-shifted)” but the formula $g_s = GM/(R^2\sqrt{1-u})$ computes the *local* (proper) surface gravity. The formula itself is correct; only the comment is misleading.
2. **Solar mass value** (`constants.jl`): Uses $M_\odot = 1.98892 \times 10^{33}$ g (older value). The IAU 2015 nominal value gives $\approx 1.98848 \times 10^{33}$ g (0.02% difference). Negligible for EOS-level calculations.
3. **Gaunt factor table dependency**: The pipeline reads `gffgu.dat` (Sutherland 1998 tabulated Gaunt factors) as a production input, violating the project’s zero-table-dependency principle. A first-principles computation from Coulomb wave functions is planned.

No discrepancies were found in any physics equations or fit parameters.

References

- Auer, L. H. 1976, JQSRT, 16, 931
- Beloborodov, A. M. 2002, ApJ, 566, L85
- Greenstein, G. & Hartke, G. J. 1983, ApJ, 271, 283
- Haakonsen, C. B., Turner, M. L., Tacik, N. A., & Rutledge, R. E. 2012, ApJ, 749, 52
- Pechenick, K. R., Ftaclas, C., & Cohen, J. M. 1983, ApJ, 274, 846
- Potekhin, A. Y., Fantina, A. F., Chamel, N., Pearson, J. M., & Goriely, S. 2013, A&A, 560, A48
- Potekhin, A. Y. & Chabrier, G. 2003, ApJ, 585, 955
- Reinhard, E., Stark, M., Shirley, P., & Ferwerda, J. 2002, ACM TOG, 21, 267
- Rybicki, G. B. 1971, JQSRT, 11, 589
- Shapiro, S. L. & Teukolsky, S. A. 1983, Black Holes, White Dwarfs, and Neutron Stars (Wiley)
- Suleimanov, V., Potekhin, A. Y., & Werner, K. 2009, A&A, 500, 891
- Sutherland, R. S. 1998, MNRAS, 300, 321